

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

11

MARKING SCHEME

JUNE 2014

CHEMISTRY 9189/5

(a) Accuracy Marking

Calculate the difference $\pm$ (titre) sup (titre) candidate and award marks as follows:

Accuracy table

Diff.	± 0.10	≤ 0.15	≤ 0.25	≤ 0.35	≤ 0.40	≤ 0.50	≤ 0.55	≤ 0.65	≤ 0.75	≤ 0.85	≤ 0.95	≤ 1.00	≤ 1.05
Mark	12	11	10	9	8	7	6	5	4	3	2	1	0

Spread Penalty Table

Difference	≤ 0.55	0.60	0.65	0.70	0.75	0.80
Mark deducted	7	8	9	10	11	12

Difference	≤ 0.20	≤ 0.25	0.35	0.40	0.45	0.50
Mark deducted	0	1	2	3	4	5

(b) Measuring cylinder should be on flat surface to avoid parallax error. [1]

(c) $n = cv$

$$= 0.02 \times \text{titre value}$$

[1]

(d) 5:2

$$\text{Ans from (b)} \times \frac{5}{2}$$

[1]

(e) $n(\text{S}_2\text{O}_3^{2-}) = 0.01 \times 0.05$

$$n(\text{H}_2\text{SO}_4) = 0.015 \times 0.1$$

[1]

react in the ratio 1:1



[1]

(f) 1:1

$$(0.01 \times 0.05) \text{ moles}$$

[12]

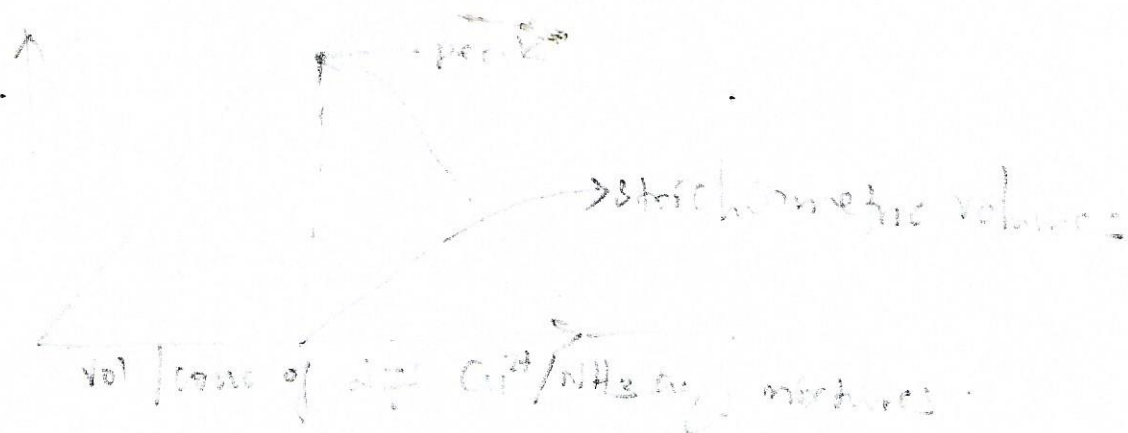
(g) Use of measuring cylinders

[1]

(h) Cl^- are oxidised by MnO_4^-

Experiment 2.1 To determine

the stoichiometry ratio of a complex ion eg $\text{Cu}^{2+}/\text{NH}_3$ eq.



Method :-

- measure the absorbance (intensities) of different volume mixtures of Cu^{2+} & NH_3 (eg $10\text{ cm}^3 \text{ Cu}^{2+} + 10\text{ cm}^3 \text{ NH}_3$ to $10\text{ cm}^3 \text{ Cu}^{2+} + 20\text{ cm}^3 \text{ NH}_3$).
- Plot a graph of absorbance vs vol / conc of diff mixtures.
- The highest point, the peak gives the equivalent reacting volumes. Calculate the ratios of the no. of moles.

Results $n(\text{Cu}^{2+}) : n(\text{NH}_3)$

$$\frac{C_1 V_1}{1000} : \frac{C_2 V_2}{1000}$$

divide by the smallest #

$$1 : 1$$

Step 1. Place b/w 30 and 60 cm³ dH₂O into a plastic cup using a measuring cylinder

Marking points: - Range of dH₂O volume 1
 - mention of apparatus used 1
 * Justification of vol used as
 - Enough to completely immerse thmt bulb
 - large enough to dissolve the solute

Step 2: Weigh a m (NaHCO₃) of b/w 1 < x < 15g into a dry plastic cup using an electronic balance

Marking points: - Range of mass used 1 (R) 1g or 15g and 'known mass'
 - mention of apparatus used (plastic cup + balance)

Justification of mass used as:

- large enough to give a reasonable ΔT
 - small enough to completely dissolve

Step 3: Insert thmt into the measured dH₂O, stir gently 1 and ~~record~~ note the constant initial temp. 1

Step 4: Record the initial Temp into a suitable table. 1

Marking points: - mention of recording
 - Specifying the table with units 1

e.g

Final Temp readings / °C	
Initial Temp readings / °C	
Temp change / °C	

Step 5. Add dH₂O at once into the weighed NaHCO₃ 1

Step 6. (A) reverse addition if it has the concept of residue
 Immediately insert a thmt, stir gently 1 and note the final temp. reached. 1

Step 7 Record the Final Temp into the table 1

Step 8 Calculate the Temp change ΔT by subtracting Initial Temp from Final Temp 1

Step 9: Calculate heat absorbed as

$$VC\Delta T = V(\text{H}_2\text{O}) \times 4.2 \times \Delta T \quad \underline{1}$$

Step 10: Calculate $n(\text{NaHCO}_3)$ used as

$$= \frac{m(\text{NaHCO}_3)}{M_r(\text{NaHCO}_3)} = \frac{5}{84} \quad \underline{1}$$

Step 11: Calculate heat of soln as

$$\frac{VC\Delta T}{n(\text{NaHCO}_3)} \quad \underline{1}$$